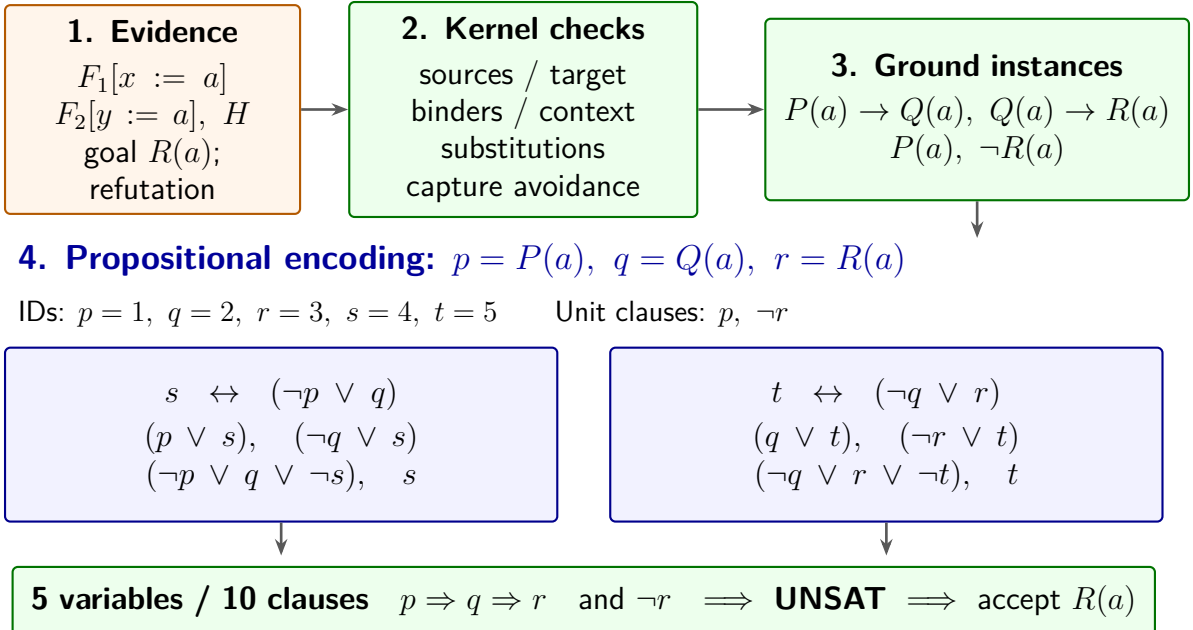

1. Evidence
$F_1[x := a]$
$F_2[y := a],\ H$
goal $R(a)$;
refutation
2. Kernel checks
sources / target
binders / context
substitutions
capture avoidance
3. Ground instances
$P(a) \to Q(a),\ Q(a) \to R(a)$
$P(a),\ \neg R(a)$
4. Propositional encoding: $p = P(a),\ q = Q(a),\ r = R(a)$
IDs: $p = 1,\ q = 2,\ r = 3,\ s = 4,\ t = 5$
Unit clauses: $p,\ \neg r$
$s \leftrightarrow (\neg p \vee q)$
$(p \vee s),\quad (\neg q \vee s)$
$(\neg p \vee q \vee \neg s),\quad s$
$t \leftrightarrow (\neg q \vee r)$
$(q \vee t),\quad (\neg r \vee t)$
$(\neg q \vee r \vee \neg t),\quad t$
5 variables / 10 clauses $p \Rightarrow q \Rightarrow r$ and $\neg r \implies$ UNSAT $\implies$ accept $R(a)$